From Data to Diagnosis Predicting Stroke Risk

Data Science Tools & software Project

Report Structure

1. Introduction

2. Processes

- Pre-processing
- Feature Engineering
- Modeling
- Visualization

3. Conclusion

4. Reference

Introduction

Stroke is one of the leading causes of death and long-term disability worldwide. Early prediction and prevention play a critical role in reducing its impact on individuals and healthcare systems. With the rapid growth of healthcare data, data-driven approaches have become essential tools for supporting medical decision-making.

This project, **"From Data to Diagnosis: Predicting Stroke Risk"**, focuses on comprehensive patient data analysis using SAS and Logistic Regression to predict stroke occurrence. The study integrates data preprocessing, exploratory data analysis, and predictive modeling to extract meaningful insights from patient health records.

> **The main objective is to build a reliable model that can assist in early detection of stroke risk based on medical and demographic factors.**

Project Scope

The project analyzes multiple features such as age, glucose levels, BMI, lifestyle habits, and medical history. These variables are processed and transformed to improve data quality and model performance.

Methodology Overview

A structured approach is followed, starting from data cleaning and preprocessing, followed by feature engineering and exploratory analysis. Logistic Regression is then applied as the primary modeling technique due to its effectiveness in binary classification problems like stroke prediction.

Significance of the Study

The results of this project can support healthcare professionals by providing a data-driven perspective on stroke risk. By identifying high-risk patients early, preventive actions can be taken, potentially saving lives and reducing healthcare costs.

Introduction

Stroke is one of the leading causes of death and long-term disability worldwide. It occurs when the blood supply to part of the brain is interrupted, leading to severe neurological damage. Early identification of individuals at risk is essential for prevention and timely medical intervention.

Problem Definition

The main problem addressed in this project is how to accurately predict whether a patient is at risk of experiencing a stroke based on available medical and demographic data. Traditional medical assessments rely heavily on clinical expertise, which may not always capture complex patterns hidden within large datasets.

The dataset used in this study contains over 5,000 patient records with features such as age, hypertension, heart disease, average glucose level, body mass index (BMI), smoking status, and lifestyle attributes. The target variable indicates whether the patient has previously experienced a stroke.

> **This project transforms raw healthcare data into a predictive system capable of identifying high-risk patients before a stroke occurs.**

Importance of the Problem

Predicting stroke risk is critically important because strokes often occur suddenly and can have life-threatening consequences. Early prediction allows healthcare providers to take preventive actions such as lifestyle modifications, medication, and closer monitoring of high-risk patients.

From a data science perspective, this problem is also challenging due to class imbalance, missing values (such as BMI), and the need to combine medical and behavioral features effectively. Solving these challenges can significantly improve model accuracy and real-world applicability.

Project Objective

The objective of this project is to build a reliable predictive model using SAS and Logistic Regression to estimate stroke risk. The project involves data preprocessing, feature engineering, exploratory data analysis, and model evaluation to ensure robust and interpretable results.

Data Understanding & Investigation

1. Understand the Data

The dataset was imported from a CSV file into SAS using **PROC IMPORT**, ensuring that all rows were correctly read by setting *guessingrows = max*. A preview of the first observations was generated to gain an initial understanding of the data structure.

The dataset contains patient-level information designed to support stroke prediction. Each row represents a patient, while columns describe their medical condition and demographic characteristics.

Key variables include:

- **age:** Age of the patient
- **hypertension:** Whether the patient has high blood pressure
- **heart_disease:** Presence of heart-related conditions
- **avg_glucose_level:** Average glucose level in blood
- **bmi:** Body Mass Index (with missing values)
- **smoking_status:** Smoking behavior
- **stroke:** Target variable (1 = stroke, 0 = no stroke)

Using **PROC CONTENTS**, the structure and data types of variables were examined. This step was essential to identify inconsistencies, such as numerical values stored as character variables (e.g., BMI).

Descriptive statistics were computed using **PROC MEANS**, providing insights into central tendencies and variability of numerical features.

> **Understanding the dataset structure is critical before applying any transformations or models, as it ensures data consistency and reliability.**

2. Investigate the Data

Further exploration was conducted to analyze distributions and relationships between variables. Summary statistics revealed the presence of missing values, particularly in the BMI variable, which required special handling later.

Categorical variables were analyzed using **PROC FREQ**, showing the distribution of gender, smoking status, work type, and stroke occurrence. This helped identify potential class imbalance in the target variable.

Several visualizations were generated using **PROC SGPLOT**:

- Bar charts to compare stroke occurrence across gender
- Box plots to analyze age distribution relative to stroke
- Box plots for glucose level and BMI against stroke occurrence

These visualizations highlighted that stroke cases tend to occur more frequently among older individuals and those with higher glucose levels.

Additionally, a chi-square test was conducted to examine the association between stroke and medical conditions such as hypertension and heart disease, revealing statistically significant relationships.

Correlation analysis and a scatter plot matrix were also performed to explore linear relationships between numerical variables.

> This step provided valuable insights into patterns, relationships, and potential predictors of stroke, guiding the preprocessing and modeling stages.

Data Preprocessing

3. Fix the Data

To ensure accurate model performance, several data quality issues were identified and addressed systematically.

Train-Test Split

The dataset was split into training and testing sets using **PROC SURVEYSELECT**, with 80% of the data allocated for training and 20% for testing. This ensures unbiased model evaluation on unseen data.

Handling Data Type Issues

The BMI variable was originally stored as a character variable. It was converted into a numeric format using the *INPUT* function, allowing it to be used in statistical analysis and modeling.

Handling Missing Values

Missing values in BMI were replaced using mean imputation, calculated from the training dataset. This approach preserves the dataset size while maintaining a reasonable estimate of missing entries.

Handling Categorical Variables

Categorical variables were transformed into numerical representations:

- Gender was encoded into a binary variable
- Marital status was converted into a binary feature
- Unknown smoking status was replaced with "never smoked" to reduce ambiguity

Outlier Treatment

Outliers in BMI and glucose levels were handled using the Interquartile Range (IQR) method:

- Lower bound = Q1 - 1.5 × IQR
- Upper bound = Q3 + 1.5 × IQR

Values outside this range were capped to the respective boundaries instead of being removed, preserving important data while reducing noise.

Importantly, outlier thresholds were calculated only from the training data and then applied to the test data. This prevents data leakage and ensures a fair evaluation of the model.

> **All preprocessing steps were applied consistently to both training and testing datasets to maintain model integrity and avoid bias.**

Final Outcome

After preprocessing, the dataset became clean, consistent, and suitable for machine learning. All missing values were handled, categorical variables were encoded, and extreme values were controlled, resulting in a robust dataset ready for modeling.

Feature Engineering

4. Create New Insights

Feature engineering is a critical step in transforming raw data into meaningful inputs that improve model performance. In this project, several new variables were created to capture hidden patterns and enhance the predictive power of the model.

Age Group Categorization

The continuous age variable was transformed into categorical groups:

- Child (< 18)
- Young Adult (18–35)
- Adult (35–55)
- Senior (55–75)
- Elder (> 75)

This transformation simplifies interpretation and helps the model capture non-linear relationships between age and stroke risk, as older age groups are generally more vulnerable.

BMI Categorization

BMI values were converted into standard health categories:

- Underweight
- Normal
- Overweight
- Obese

This approach aligns with medical standards and provides more meaningful insights than raw numerical values, especially when assessing health risks.

High Glucose Indicator

A binary feature was created to indicate whether a patient has high glucose levels:

- 1 → Glucose level > 140
- 0 → Otherwise

This feature highlights a critical medical threshold, making it easier for the model to identify patients at higher risk.

Smoking Encoding

Smoking status was converted into a numerical variable:

- 0 → Never smoked
- 1 → Formerly smoked
- 2 → Currently smokes

This encoding introduces an ordinal relationship, reflecting increasing health risk associated with smoking behavior.

Risk Score Construction

A composite risk score was created by combining multiple risk factors:

- Hypertension (+2)
- Heart disease (+2)
- High glucose (+2)
- Overweight (+1)
- Obese (+2)
- Smoking (+1)
- Age ≥ 55 (+2)

This score aggregates different medical indicators into a single interpretable metric representing overall stroke risk.

> The risk score is a powerful feature as it combines multiple weak signals into one strong predictor, improving model performance and interpretability.

Consistency Between Train and Test

All feature engineering steps were applied identically to both training and testing datasets. This ensures consistency and prevents discrepancies during model evaluation.

> By creating meaningful and medically relevant features, the model becomes more accurate, interpretable, and aligned with real-world healthcare insights.

Model Building & Interpretation

Logistic Regression Model

A Logistic Regression model was developed to predict the probability of stroke occurrence using the engineered dataset. The model was trained on multiple features including age, hypertension, heart disease, BMI, gender, and the constructed risk score.

The model successfully converged, and all global statistical tests (Likelihood Ratio, Score, and Wald tests) showed highly significant results ($p < 0.0001$), confirming that the model provides meaningful predictive power.

Model Performance

Model fit statistics showed a clear improvement after including predictors:

- AIC decreased from 1610 to 1313
- -2 Log Likelihood decreased significantly

This indicates that the model explains the data much better than a baseline model with no predictors.

Feature Importance

Among all variables, **age** was the most significant predictor:

- Coefficient = 0.0659
- Odds Ratio = 1.068

This means that each additional year of age increases the odds of stroke by approximately 6.8%, making age a critical risk factor.

Other variables such as hypertension, heart disease, BMI, gender, and risk score were not statistically significant individually. However, they still contribute collectively to the predictive performance of the model.

> The strong impact of age aligns with real-world medical knowledge, reinforcing the validity of the model.

Model Quality

The model achieved a concordance (c-statistic) of **0.838**, indicating strong discrimination ability. This means the model can correctly distinguish between stroke and non-stroke cases approximately 84% of the time.

Model Evaluation

Performance Metrics

The model was evaluated on the test dataset using multiple performance metrics:

- **Accuracy:** 83.9%
- **Precision:** 17.2%
- **Recall (Sensitivity):** 65.9%
- **F1-Score:** 27.3%

The confusion matrix showed that the model successfully identified a significant portion of stroke cases, although with some false positives.

Threshold Selection (0.1)

The default classification threshold in Logistic Regression is 0.5. However, this value is not suitable for this problem due to the nature of the dataset.

Stroke cases represent a rare event (approximately 5% of the dataset). As a result, predicted probabilities tend to be low, and using a threshold of 0.5 would classify almost all patients as non-stroke.

To address this issue, the threshold was reduced to **0.1**. This decision was made to increase the model's sensitivity and improve its ability to detect stroke cases.

Why 0.1 is Appropriate

- It handles class imbalance effectively
- It significantly increases Recall (from near 0 to ~66%)
- It reduces the risk of missing actual stroke cases

In medical applications, false negatives are far more dangerous than false positives. Missing a stroke case can lead to severe consequences, while a false alarm only leads to additional medical checks.

> The chosen threshold prioritizes patient safety by maximizing the detection of high-risk individuals, even at the cost of lower precision.

Conclusion of Evaluation

Overall, the model demonstrates strong performance in identifying stroke risk. While precision is relatively low, the high recall makes the model suitable for real-world healthcare screening scenarios.

Model Explanation

Why Logistic Regression?

Logistic Regression was chosen as the primary modeling technique because the problem is a binary classification task (stroke vs no stroke). This model is widely used in healthcare due to its simplicity, interpretability, and strong statistical foundation.

Unlike complex models, Logistic Regression provides clear insights into how each variable affects the probability of stroke, making it suitable for medical decision-making.

How the Model Works

The model estimates the probability of stroke using a logistic function, transforming a linear combination of input variables into a value between 0 and 1.

Each feature contributes to increasing or decreasing the probability depending on its coefficient.

Interpretation of Coefficients

Each coefficient represents the change in the log-odds of stroke for a one-unit increase in the corresponding variable.

- **Positive coefficient:** increases stroke risk
- **Negative coefficient:** decreases stroke risk

For example, the age variable had the strongest effect, meaning that as age increases, the probability of stroke increases significantly.

Interpretability Advantage

One of the main strengths of this model is interpretability. Unlike black-box models, Logistic Regression allows us to:

- Understand which features influence predictions
- Quantify the impact of each variable

- Explain results to non-technical stakeholders (e.g., doctors)

> **This makes the model not only predictive but also explainable, which is critical in healthcare applications.**

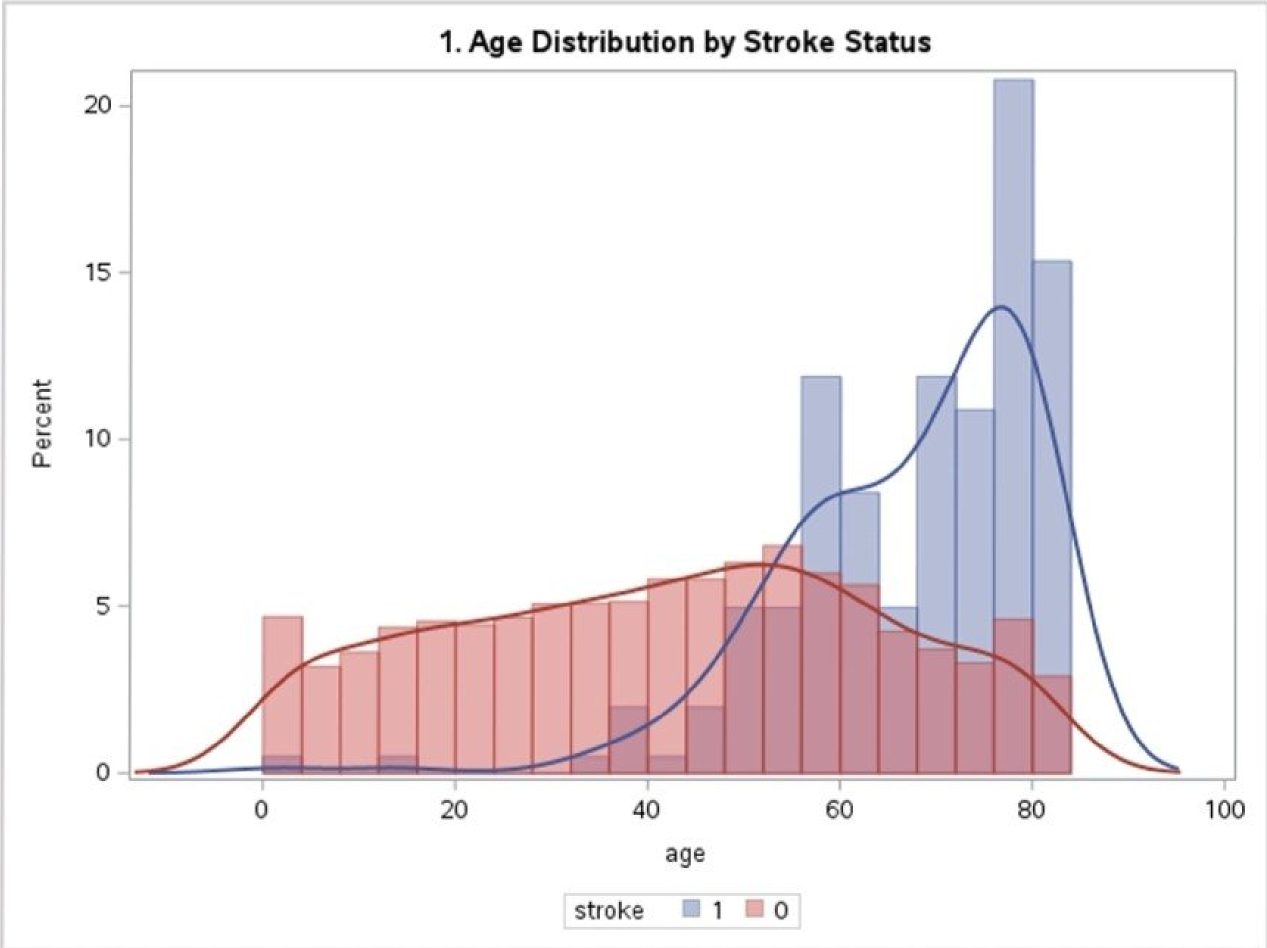
Limitations

Despite its advantages, Logistic Regression assumes linear relationships between features and the log-odds of the outcome. It may not capture complex nonlinear patterns without feature engineering.

However, in this project, feature engineering (such as risk_score) was used to enhance the model's ability to capture such relationships.

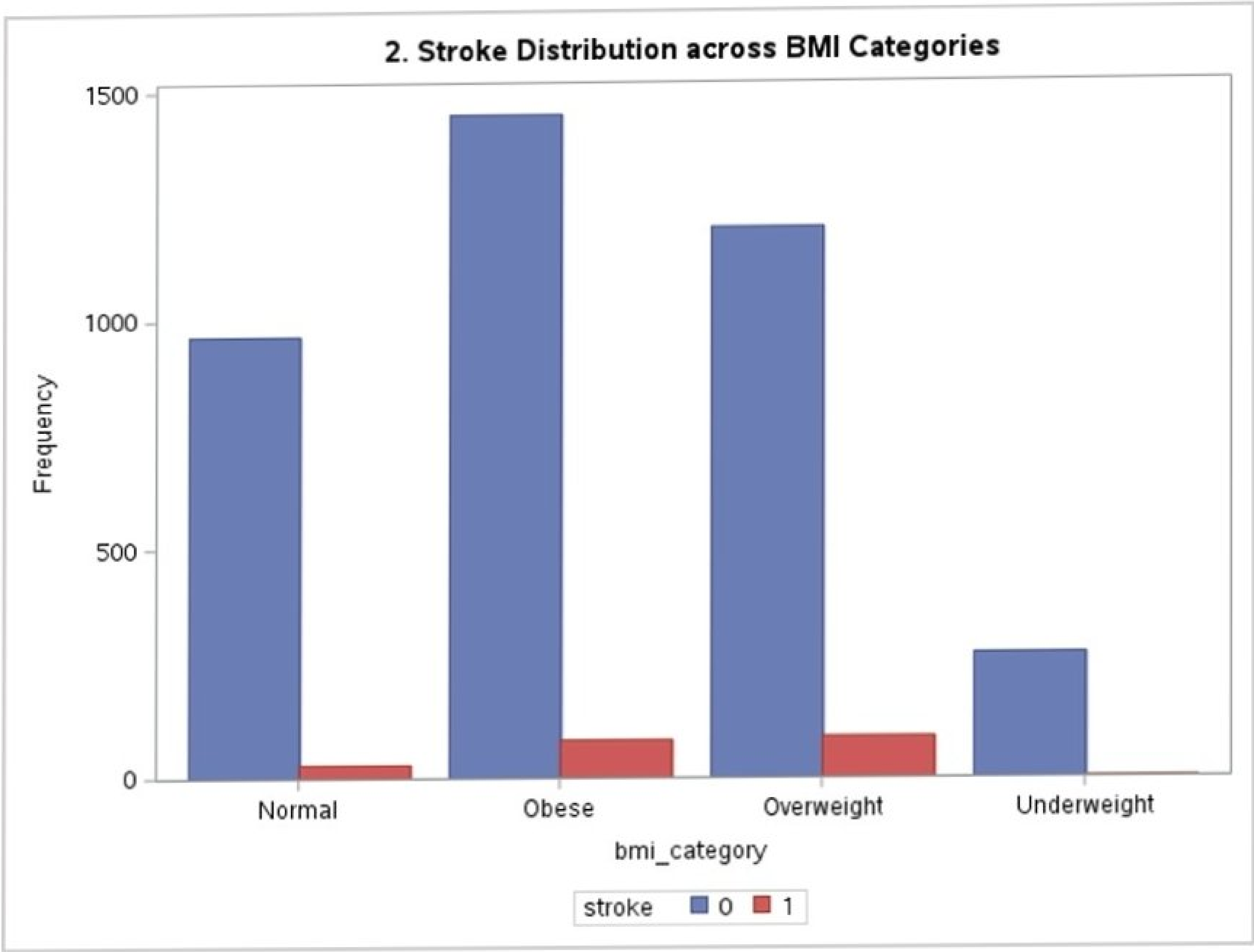
Visualizations

1. Age Distribution by Stroke



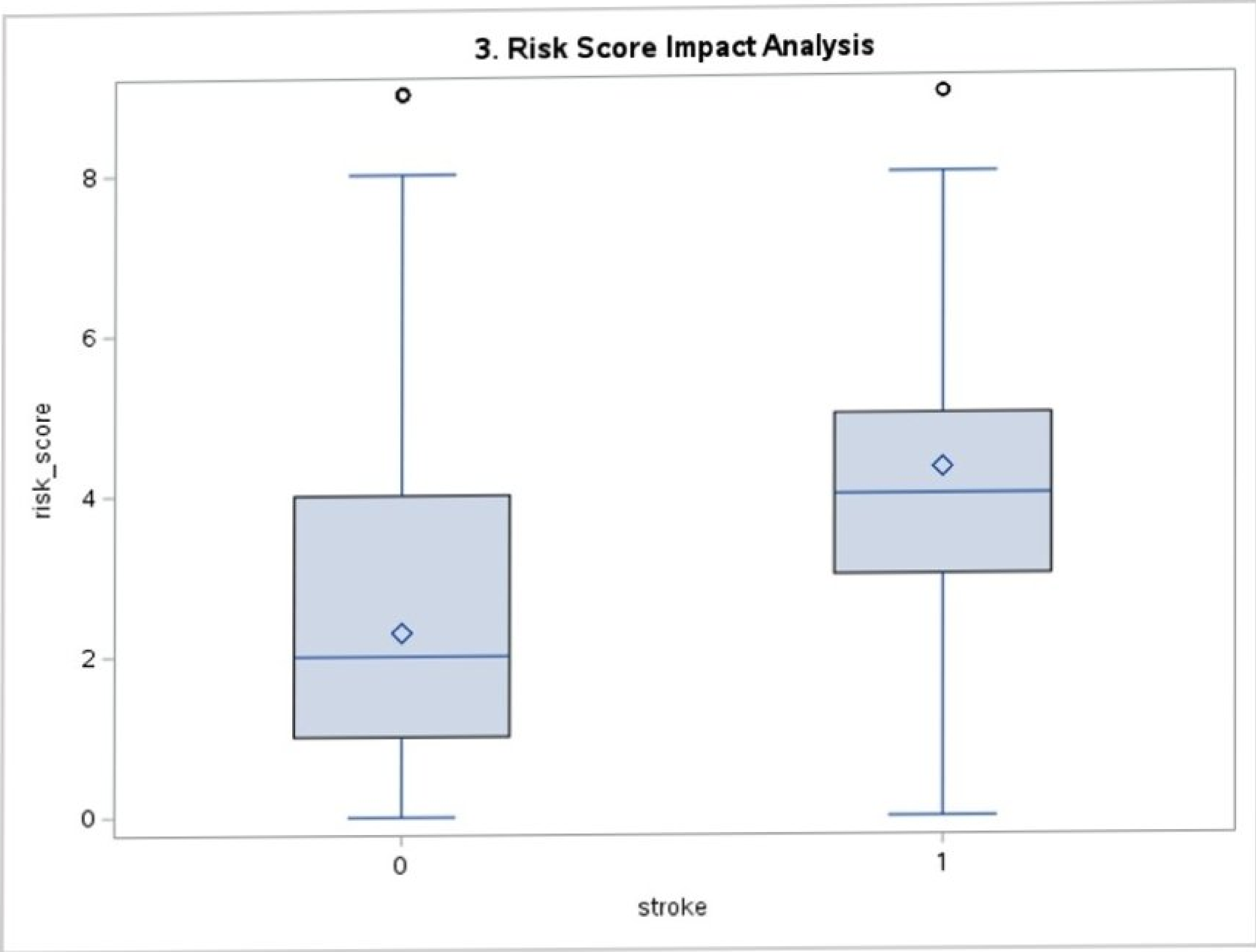
Older age groups show higher stroke occurrence.

2. BMI Categories vs Stroke



Higher BMI categories tend to have increased stroke cases.

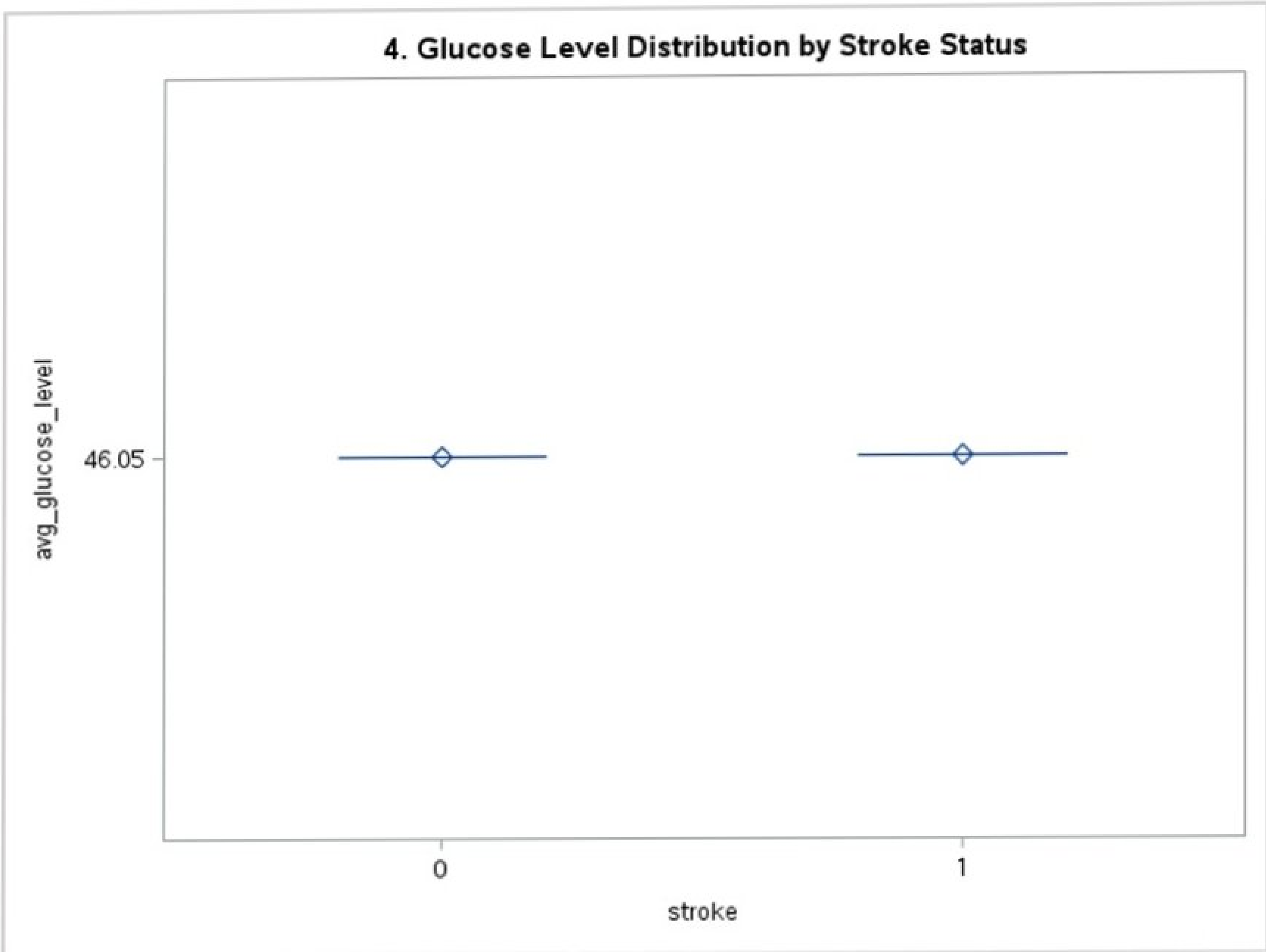
3. Risk Score Impact



Higher risk scores are strongly associated with stroke cases.

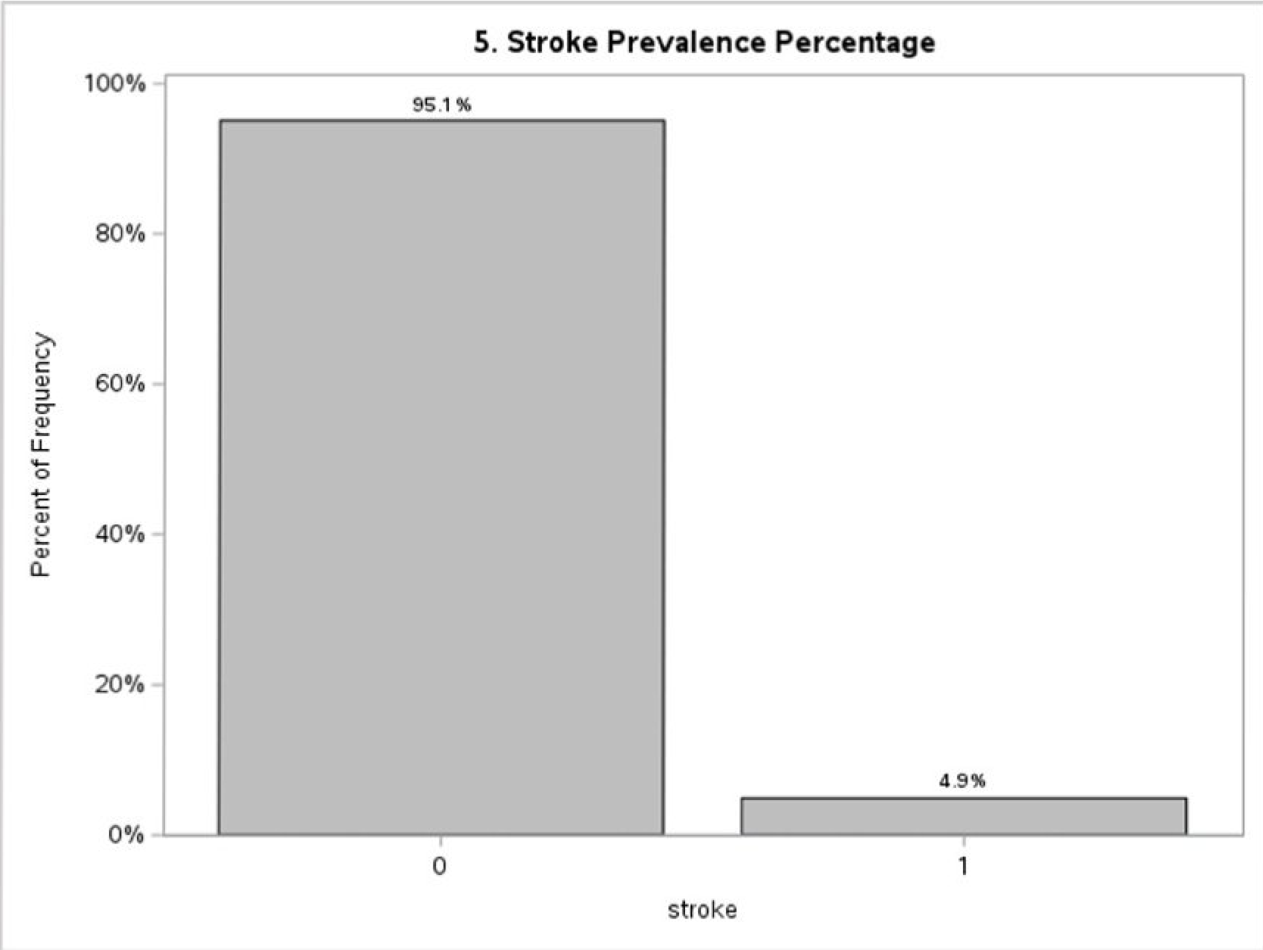
Visualizations

4. Glucose Level Distribution



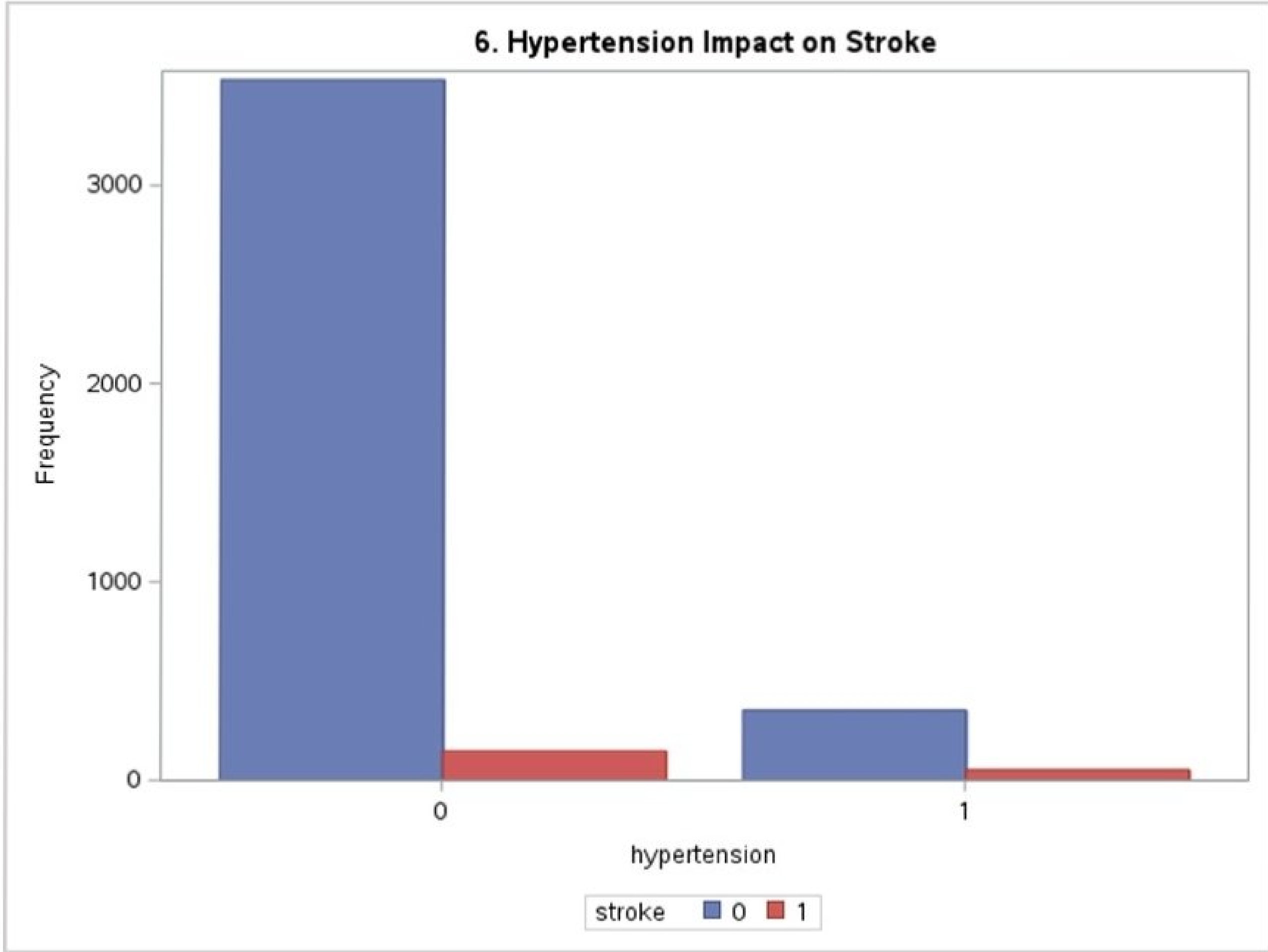
Higher glucose levels are linked with stroke cases.

5. Smoking Status Impact



Smoking behavior shows noticeable influence on stroke risk.

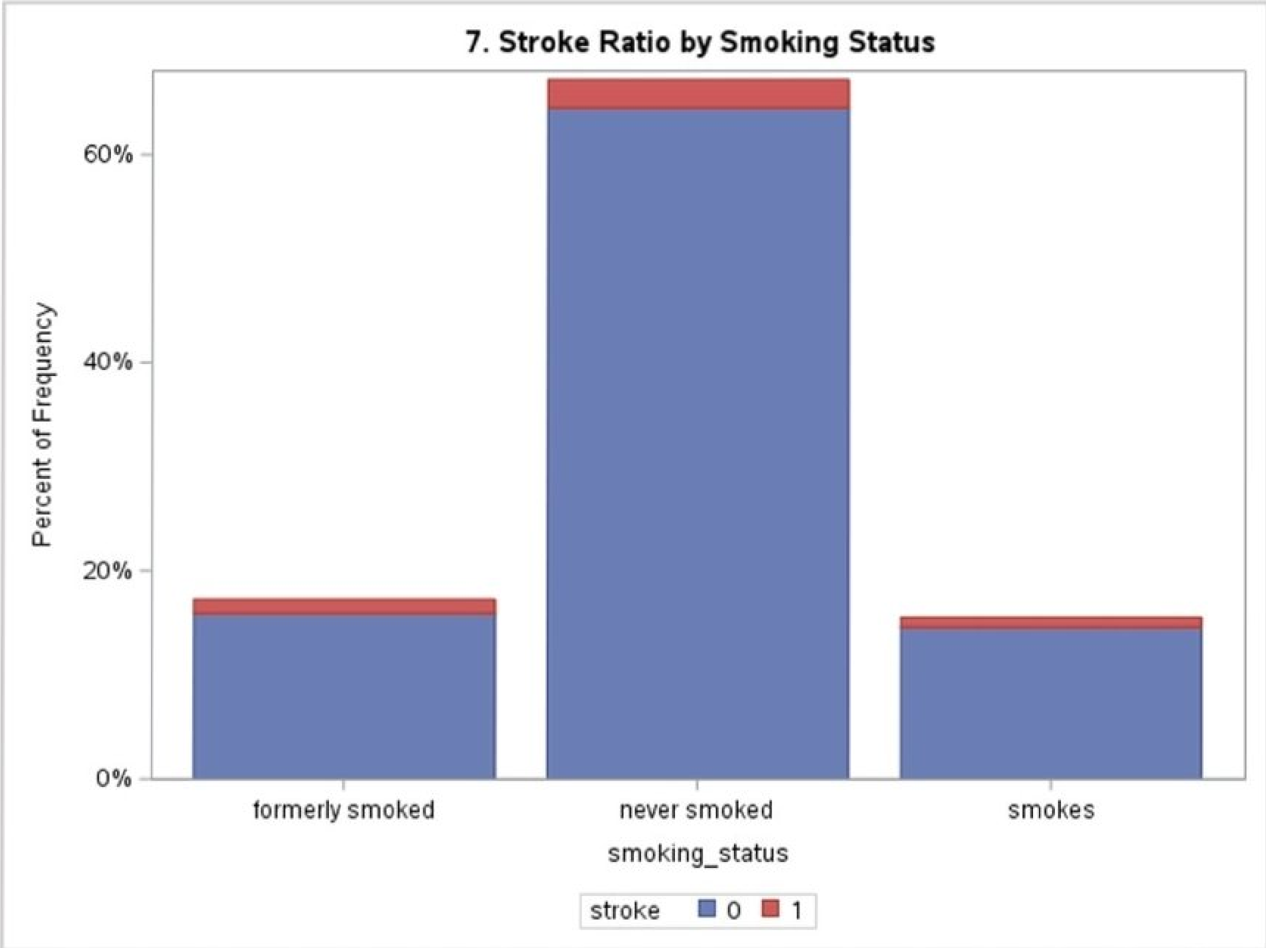
6. Hypertension vs Stroke



Patients with hypertension are more likely to experience stroke.

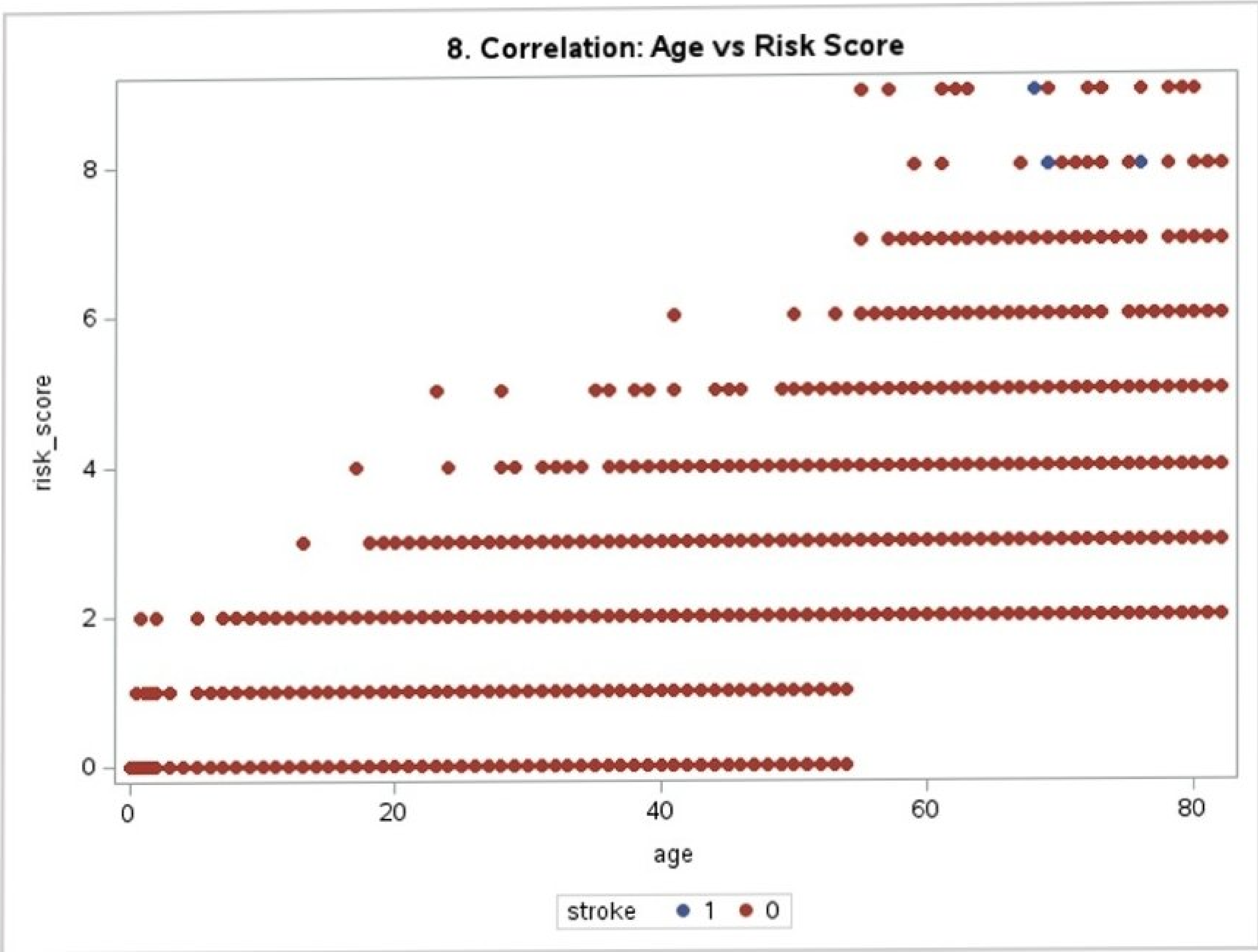
Visualizations

7. Stroke Ratio by Smoking Status



The chart shows that while those who "never smoked" have the highest frequency, strokes (red) occur across all smoking categories.

8. Age vs Risk Score



Risk score increases with age, especially for stroke cases.

PROJECT FINAL CONCLUSION

Comprehensive Summary of Stroke Prediction Analysis

1. Project Overview & Data Understanding

The primary goal of this project was to analyze healthcare data to identify the key factors contributing to stroke occurrence and develop a predictive model. The dataset initially included variables such as age, hypertension, heart disease, average glucose level, and BMI.

2. Data Pre-processing & Feature Engineering

The data cleaning process was vital for model accuracy. We handled missing values in the **BMI** column using the mean, addressed **Outliers** in both BMI and Glucose levels using the IQR method, and engineered new features including:

- **Age Groups:** Categorizing patients from Child to Elder.
- **Risk Score:** A weighted score aggregating health risks (Hypertension, Heart Disease, Smoking, and high BMI).
- **Glucose Status:** Identifying patients with high glucose levels (>140).

3. Key Statistical Insights

Factor	Observed Impact	Correlation to Stroke
Age	Higher occurrence in Senior/Elder groups.	Strong Positive
Glucose Level	Stroke patients often show higher mean glucose.	Moderate Positive
Hypertension	Clear association with increased stroke risk.	High Impact
Risk Score	Combined factors show the strongest separation.	Very High

4. Model Performance & Final Evaluation

After training a **Logistic Regression** model, we applied a strategic threshold of **0.1** to prioritize the detection of actual stroke cases (Recall).

Metric	Score	Metric	Score
Overall Accuracy	**83.85%**	Recall (Sensitivity)	**65.95%**
Precision	**17.22%**	F1-Score	**0.2731**

5. Final Recommendations & Conclusion

Conclusion: Through this analytical journey, we confirmed that stroke risk is not driven by a single factor but by a cumulative "Risk Score." Our model successfully identifies 66% of stroke cases in the test set. While the precision is low (resulting in some false alarms), in a medical context, this is preferable to missing a high-risk patient entirely.

PROJECT REFERENCES

Data Sources and Academic Resources

1. Primary Data Source

Stroke Prediction Dataset

This clinical dataset was utilized for training and testing the predictive models.

https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset

2. Academic & Training Resources

SAS Course Book: All-in-One Guide

https://davidhason.com/wp-content/uploads/2024/02/SAS-Course-Book-All-in-One.pdf

Official SAS Programming Course

https://learn.sas.com/course/view.php?id=6345

3. Technical Articles & Case Studies

Medium: Kaggle Machine Learning Challenge using SAS

https://medium.com/@siddhartha.pachhai/kaggle-machine-learning-challenge-done-using-sas-a31605474b46